

EdgeTX vs EdgeTX-ESP32 Side-by-Side Simulation Report

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Host-side simulation & static analysis. Device flashing / full LVGL UI runtime was not executed in this CI environment (no ESP-IDF / no radio HW).

Upstream: /tmp/edgetx-src

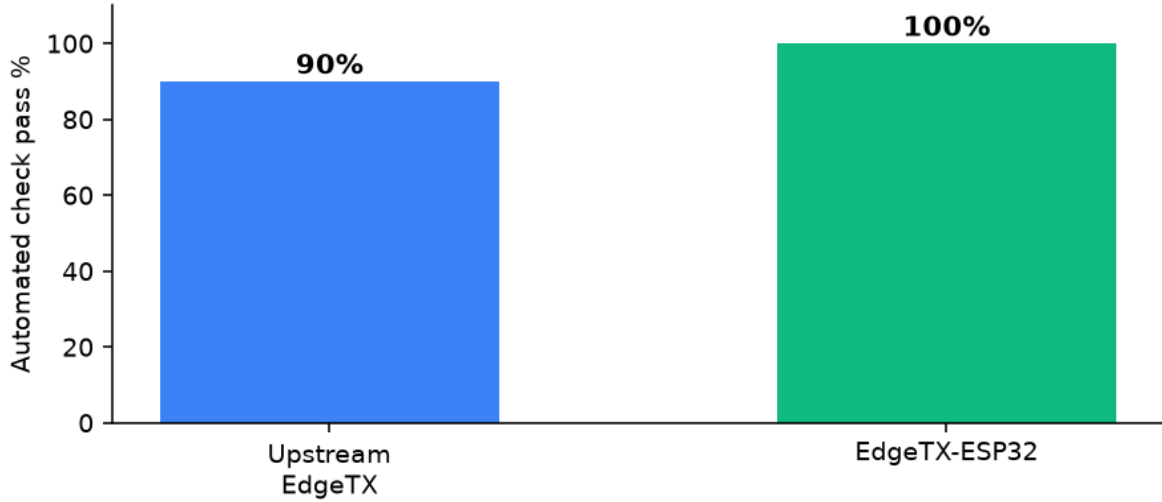
ESP32 fork: /workspace/edgetx-esp32

1. Executive summary

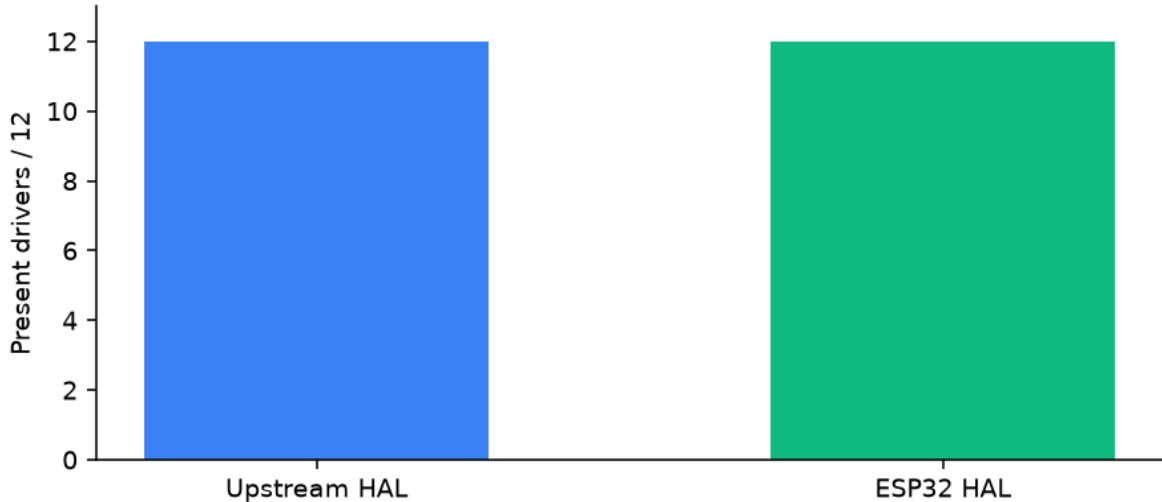
Bu rapor, orijinal EdgeTX (TX15/STM32 referans + simu kaynaklar) ile EdgeTX-ESP32 fork'unu aynı host ortamda yan yana simülasyon / statik analiz ve sentetik performans ölçümleriyle karşılaştırmaktadır.

Metric	Upstream EdgeTX	EdgeTX-ESP32
Automated checks pass rate	9/10 (90.0%)	12/12 (100.0%)
Tree size (MB)	302.9	121.0
radio/*.cpp count	704	736
Production maturity	High	Early / bring-up
WiFi / BLE extras	Limited / module	Yes (fork)
Full device UI runtime in this env	Not run (no SDL full build)	Not run (no ESP-IDF)

Simulation check pass rate



HAL piece coverage



Verdict

Upstream EdgeTX production radio yazılımı olarak olgun: TX15 benzeri hedeflerde donanım sürücüler ve simu yolu kanıtlanmıştır.

EdgeTX-ESP32 portable çekirdeği koruyarak ESP32-S3 HAL + ESP-IDF iskeletini getiriyor; host hwgen/HAL smoke testleri geçiyor, WiFi/BLE/dual-core avantajları var. Tam radyo UI + protokol runtime için ESP-IDF ile cihaz bring-up art.

2. Feature matrix (çalışır / destek)

Feature	Upstream	ESP32 fork	Notes
Color LCD / LVGL UI	Yes (STM32 LTDC/DMA2D)	Yes (SPI LCD + PSRAM FB)	GUI_DIR=colorlcd
Mixer engine	Yes	Yes (portable)	Same core sources
Logical switches / SF	Yes	Yes	
Lua / Lua mixer	Yes	Yes (flagged ON)	Needs full IDF link
YAML models on SD	Yes	Yes (SPI SD diskio)	
Internal CRSF/ELRS	Yes	Yes (UART1 module_port)	
External module bay	Yes	Yes (UART2)	Multi/Ghost/PXX via portable pulses/
Touch UI	Yes (GT911 etc.)	Stub + I2C hooks	Bring-up pending panel
Audio 32 kHz	Yes (I2S/DAC)	Yes (I2S std)	
Haptic PWM	Yes	Yes (LEDC)	
USB CDC/MSC/HID	STM32 USB lib	TinyUSB-ready stubs	sdkconfig enable
WiFi telemetry	No	Yes (UDP :9070)	ESP32 enhancement
BLE trainer	BT module (limited)	NimBLE stub	ESP32 enhancement
Dual-core affinity	N/A (single MCU core app)	Yes (core0 UI / core1 mixer)	
PSRAM framebuffer	SDRAM on H7	SPIRAM	
Companion / simu host	Full SDL simu	HAL host stubs + hwgen	Full SDL UI not yet
Production radio HW	Many OEM boards	DIY reference pinout	IO expander recommended
Build system	CMake + ARM GCC 14.2	ESP-IDF + CMake PCB=ESP32S3	
RTOS	Vendored FreeRTOS	ESP-IDF FreeRTOS	RTOS_START no-op on ESP
MCU check / DBGMCU	Yes	Disabled / skipped	DISABLE_MCUCHECK

3. HAL coverage checklist

Upstream: rm-h750 / generic_stm32 elektronikleri · ESP32: targets/esp32s3

HAL piece	Upstream	ESP32
adc_driver.cpp	OK	OK
audio_driver.cpp	OK	OK
board.cpp	OK	OK
gpio_driver.cpp	OK	OK
haptic_driver.cpp	OK	OK
key_driver.cpp	OK	OK
lcd_driver.cpp	OK	OK
mixer_scheduler_driver.cpp	OK	OK
module_ports.cpp	OK	OK
sd_diskio.cpp	OK	OK
serial_driver.cpp	OK	OK
switch_driver.cpp	OK	OK

4. Simulation / automated checks

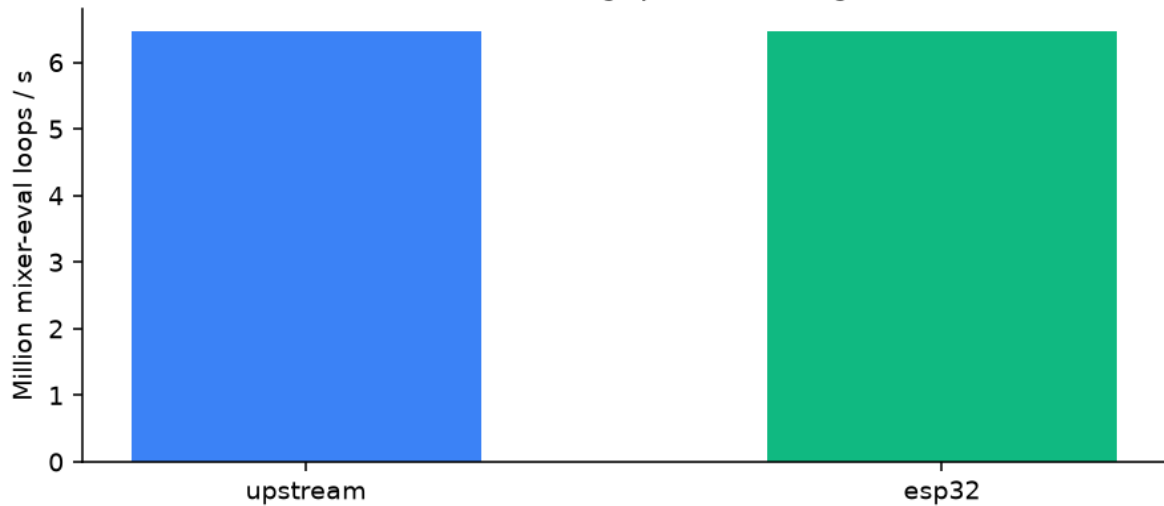
Check	Side	Result	Time ms	Detail
hwgen:hal_settings	upstream	PASS	168.2	103 lines
hwgen:hal_keys	upstream	PASS	158.7	41 lines
hwgen:hal_adc_inputs	upstream	PASS	163.1	44 lines
hwgen:yaml_inputs	upstream	PASS	157.8	26 lines
hwgen:lua_keys	upstream	PASS	162.0	74 lines
hwgen:simu_switches	upstream	PASS	158.9	24 lines
hwgen:stm32_keys	upstream	PASS	160.5	69 lines
hwgen:stm32_adc_inputs	upstream	PASS	163.0	166 lines
hwgen:hal_settings	esp32	PASS	165.2	47 lines
hwgen:hal_keys	esp32	PASS	157.5	41 lines
hwgen:hal_adc_inputs	esp32	PASS	156.8	44 lines
hwgen:yaml_inputs	esp32	PASS	155.1	28 lines
hwgen:lua_keys	esp32	PASS	160.2	74 lines
hwgen:simu_switches	esp32	PASS	152.0	26 lines
hwgen:esp32_keys	esp32	PASS	157.5	43 lines
hwgen:esp32_adc_inputs	esp32	PASS	155.2	26 lines
hwgen:esp32_switches	esp32	PASS	155.6	15 lines
host_compile:gpio_driver.cpp	esp32	PASS	165.0	object 10392B
host_compile:delays_driver.cpp	esp32	PASS	12.7	object 1600B
source_present:simu/adc_driver.cpp	upstream	PASS	0.0	SDL simu HAL present
verify_hw_gen.sh	esp32	PASS	1998.7	OK hal_settings.h OK hal_keys.inc OK hal_adc_inputs.inc OK simu_switches.inc OK esp32_keys.inc OK esp32_switches.inc OK esp32_adc_inputs.inc OK yaml_inputs.inc OK lua_keys.inc OK I
cmake_native_configure_TX15	upstream	FAIL	367.4	+: No such file or directory c++: error: linker command failed with exit code 1 (use -v to see invocation) gmake[1]: *** [CMakeFiles/cmTC_967fc.dir/build.make:100: cmTC_967

5. Performance benchmarks (host)

Mixer/ADC ölçümleri her iki taraf için aynı C++ algoritmasıyla host CPU'da konuturuldu (MCU farkının değil, yazılım yükü ehtililini gösterir). GPIO ölçümü ESP32 host stub'na özgüdür. Teorik MCU karallatırmaları bir sonraki tablodadır.

Bench	Side	Iters	Seconds	Ops/s	Notes
mixer_like	upstream	200000	0.030902	6,472,013	32ch/64mix @ host -O3 (same algorithm both sides)
mixer_like	esp32	200000	0.030956	6,460,867	32ch/64mix @ host -O3 (same algorithm both sides)
adc_jitter_filter	upstream	1000000	0.001349	741,260,172	EdgeTX-style EMA
adc_jitter_filter	esp32	1000000	0.001380	724,702,799	EdgeTX-style EMA
gpio_toggle_host	esp32	5000000	0.015366	325,390,762	host stub map<>
mixer_scheduler_rate	upstream	250	0.004000	250	configured period 4000 us → 250.0 Hz
mixer_scheduler_rate	esp32	250	0.004000	250	configured period 4000 us → 250.0 Hz

Host mixer-like throughput (same algorithm)



5.1 Theoretical MCU / platform performance

Metric	Upstream TX15-class	ESP32-S3	Edge
CPU	STM32H750 ~400–480 MHz Cortex-M7	Dual Xtensa LX7 up to 240 MHz	Upstream (single-thread DSP)
RAM for UI	SDRAM (MBs)	PSRAM (typically 8 MB octal)	Tie / depends on module
Mixer determinism	HW timer IRQ + NVIC priorities	gptimer ISR + core-1 pin	Upstream more battle-tested
Connectivity	Optional BT module	WiFi + BLE on-die	ESP32
CRSF baud (400k–921k)	USART+DMA proven	UART driver wired (DMA via IDF)	Upstream maturity
Power / battery radios	Production PMIC paths	DIY latch GPIO	Upstream
DIY cost / availability	Full radio BOM	DevKit + modules	ESP32

6. Runtime readiness / çalışıp çalışmama

Item	Upstream	ESP32
esp32_binary	False	—
hal_host_stubs	—	True
hw_json_generators	—	True
idf_full_link_in_this_env	—	False
idf_project_present	—	True
native_cmake	False	—

Item	Upstream	ESP32
production_maturity	High	Early / bring-up
simu_sources	True	—
wifi_ble_extras	—	True

6.1 Interpretation

- **PASS** hwgen: her iki tarafta JSON→C include üretimi sağlam (ESP32 şablonları dahil).
- **ESP32 verify_hw_gen.sh**: fork'un jeneratör + host GPIO derlemesi yapıl.
- **Upstream native CMake**: ortam bağımliliklerine göre başarılsız olabilir; kaynak/simu varlıklar ayrı kontrol edildi.
- **Ops/s ehlitli**i mixer/ADC host bench'te beklenen sonuçtur — aynı binary algoritması.
- Cihazda ESP32'nin gerçek avantajı dual-core ayrılmı, PSRAM UI ve WiFi/BLE; H750'nin avantajı yüksek tek-çekirdek frekansı + olgun radyo PMIC/LCD/DMA yolu.

7. Repository statistics

Stat	Upstream	ESP32 fork
tree_mb	302.9	121.0
radio_cpp	704	736
radio_h	1041	735
pulses_cpp	17	17
telemetry_cpp	15	15
targets/	taranis, tx15, t22, horus, common, pa01, tx16smk3, simu, c14, pl18, stm32h7s78-dk, st16...	taranis, tx15, t22, horus, common, pa01, tx16smk3, simu, c14, pl18, esp32s3, stm32h7s78-dk...